

Motion (9th standard)

1. Which of the following is a scalar quantity?

- a) Displacement
- b) Velocity
- c) Speed
- d) Acceleration

Correct answer: c) Speed

2. Distance is a _____ quantity.

- a) vector
- b) scalar
- c) tensor
- d) none

Correct answer: b) scalar

3. The SI unit of velocity is _____.

- a) m/s
- b) km/h
- c) cm/s
- d) all

Correct answer: a) m/s

4. A body moves 200 m in 40 s. Its average speed is _____.

- a) 5 m/s
- b) 8 m/s
- c) 10 m/s
- d) 2 m/s

Correct answer: a) 5 m/s

5. Acceleration is defined as change in _____ per unit time.

- a) speed
- b) velocity
- c) distance

d) time

Correct answer: b) velocity

6. Retardation is also called _____.

a) positive acceleration

b) negative acceleration

c) zero acceleration

d) constant acceleration

Correct answer: b) negative acceleration

7. First equation of motion: $v =$ _____

a) $u + at$

b) $ut + \frac{1}{2}at^2$

c) $v^2 - u^2 = 2as$

d) $s = ut$

Correct answer: a) $u + at$

8. Second equation of motion is _____.

a) $v = u + at$

b) $s = ut + \frac{1}{2}at^2$

c) $v^2 = u^2 + 2as$

d) all

Correct answer: b) $s = ut + \frac{1}{2}at^2$

9. Third equation of motion: _____

a) $v = u + at$

b) $s = ut + \frac{1}{2}at^2$

c) $v^2 = u^2 + 2as$

d) $s = vt$

Correct answer: c) $v^2 = u^2 + 2as$

10. In uniform motion, acceleration is _____.

a) positive

b) negative

c) zero

d) infinite

Correct answer: c) zero

11. Slope of distance-time graph gives _____.

a) acceleration

b) velocity

c) speed

d) displacement

Correct answer: c) speed

12. Slope of velocity-time graph represents _____.

a) speed

b) distance

c) acceleration

d) time

Correct answer: c) acceleration

13. Area under velocity-time graph gives _____.

a) acceleration

b) displacement

c) speed

d) force

Correct answer: b) displacement

14. In circular motion, velocity is _____.

a) constant

b) zero

c) changing direction

d) both a and c

Correct answer: d) both a and c

15. Centripetal acceleration acts _____.

a) outward

b) inward

c) tangential

d) parallel

Correct answer: b) inward

16. A car accelerates from 10 m/s to 30 m/s in 5 s. Acceleration is _____.

a) 2 m/s^2

b) 4 m/s^2

c) 6 m/s^2

d) 8 m/s^2

Correct answer: b) 4 m/s^2

17. If a body covers equal distances in equal intervals of time, it has _____.

a) uniform velocity

b) non-uniform velocity

c) zero velocity

d) constant acceleration

Correct answer: a) uniform velocity

18. Displacement is a _____ quantity.

a) scalar

b) vector

c) both

d) none

Correct answer: b) vector

19. Average velocity = total _____ / total time.

a) speed

b) acceleration

c) displacement

d) distance

Correct answer: c) displacement

20. In velocity-time graph, horizontal line indicates _____.

a) zero acceleration

b) constant acceleration

c) increasing acceleration

d) decreasing acceleration

Correct answer: a) zero acceleration

21. The rate of change of displacement is _____.

a) speed

b) velocity

c) acceleration

d) distance

Correct answer: b) velocity

22. A train starts from rest and accelerates uniformly at 2 m/s^2 for 10 s. Distance covered?

a) 50 m

b) 100 m

c) 200 m

d) 10 m

Correct answer: b) 100 m

23. Uniform circular motion has _____ acceleration.

a) zero

b) constant speed

c) centripetal

d) tangential

Correct answer: c) centripetal

24. Vector quantity has both magnitude and _____.

a) time

b) direction

c) speed

d) mass

Correct answer: b) direction

25. If velocity-time graph is a straight line parallel to time axis, acceleration is _____.

a) increasing

b) decreasing

- c) zero
- d) infinite

Correct answer: c) zero

26. A body thrown upwards has acceleration due to _____.

- a) gravity
- b) friction
- c) air resistance
- d) all

Correct answer: a) gravity

27. Distance-time graph for uniform motion is _____.

- a) curved
- b) straight line
- c) parabola
- d) hyperbola

Correct answer: b) straight line

28. Deceleration means acceleration is _____.

- a) positive
- b) negative
- c) zero
- d) constant

Correct answer: b) negative

29. SI unit of acceleration is _____.

- a) m/s
- b) m/s^2
- c) km/h
- d) cm/s^2

Correct answer: b) m/s^2

30. In non-uniform motion, speed/velocity _____.

- a) constant
- b) changes

- c) zero
- d) infinite

Correct answer: b) changes

31. A scooter travels 20 km in first hour, 40 km in next hour. Average speed?

- a) 20 km/h
- b) 30 km/h
- c) 40 km/h
- d) 60 km/h

Correct answer: b) 30 km/h

32. Velocity-time graph for free fall is _____.

- a) straight line
- b) horizontal
- c) curved
- d) vertical

Correct answer: a) straight line

33. Centripetal force is provided by _____.

- a) gravity
- b) tension
- c) friction
- d) all possible

Correct answer: d) all possible

34. If $u=0$, then from $v^2 = u^2 + 2as$, $v =$ _____.

- a) at
- b) $\sqrt{2as}$
- c) $\frac{1}{2}at^2$
- d) ut

Correct answer: b) $\sqrt{2as}$

35. Motion along a straight line is _____.

- a) circular
- b) rectilinear

c) rotational

d) oscillatory

Correct answer: b) rectilinear

36. The quantity that has only magnitude is _____.

a) velocity

b) displacement

c) speed

d) force

Correct answer: c) speed

37. Area under distance-time graph gives _____.

a) velocity

b) acceleration

c) distance

d) time

Correct answer: c) distance

38. A body moving with constant velocity has _____.

a) acceleration

b) no acceleration

c) varying speed

d) retardation

Correct answer: b) no acceleration

39. In circular motion, distance travelled is _____.

a) radius

b) circumference

c) diameter

d) arc length

Correct answer: b) circumference

40. Acceleration due to gravity is approximately _____.

a) 9.8 m/s^2

b) 10 m/s^2

c) 8 m/s^2

d) 5 m/s^2

Correct answer: a) 9.8 m/s^2

41. Slope of speed-time graph gives _____.

a) velocity

b) acceleration

c) distance

d) displacement

Correct answer: b) acceleration

42. If $a = 0$, motion is _____.

a) accelerated

b) retarded

c) uniform

d) non-uniform

Correct answer: c) uniform

43. Displacement can be _____.

a) positive

b) negative

c) zero

d) all

Correct answer: d) all

44. A ball is dropped from 20 m height. Time to reach ground ($g=10 \text{ m/s}^2$)?

a) 1 s

b) 2 s

c) 4 s

d) 0.5 s

Correct answer: b) 2 s

45. Uniformly accelerated motion has constant _____.

a) speed

b) velocity

c) acceleration

d) distance

Correct answer: c) acceleration

46. Vector addition follows _____.

b) triangle law

c) parallelogram law

d) both b and c

Correct answer: d) both b and c

47. In velocity-time graph, acceleration = _____.

a) slope

b) area

c) intercept

d) curve

Correct answer: a) slope

48. A car stops from 20 m/s in 4 s. Retardation?

a) 5 m/s^2

b) -5 m/s^2

c) 4 m/s^2

d) -4 m/s^2

Correct answer: b) -5 m/s^2

49. Motion of moon around earth is _____.

a) rectilinear

b) circular

c) random

d) oscillatory

Correct answer: b) circular

50. Average speed for journey with equal time at speeds v_1 , v_2 is _____.

a) $(v_1 + v_2)/2$

b) $2v_1v_2/(v_1+v_2)$

c) v_1v_2

d) none

Correct answer: b) $2v_1v_2/(v_1+v_2)$

51. Distance-time graph for rest is _____.

a) straight line parallel to time axis

b) vertical line

c) horizontal line

d) curved

Correct answer: a) straight line parallel to time axis

52. If displacement is zero, average velocity is _____.

a) positive

b) negative

c) zero

d) infinite

Correct answer: c) zero

53. Tangential acceleration changes _____.

a) speed

b) direction

c) both

d) none

Correct answer: a) speed

54. In projectile motion, path is _____.

a) straight

b) parabolic

c) circular

d) elliptical

Correct answer: b) parabolic

55. A train 100 m long passes a pole in 5 s. Speed?

a) 10 m/s

b) 20 m/s

c) 50 m/s

d) 5 m/s

Correct answer: b) 20 m/s

56. Velocity is a _____ quantity.

a) scalar

b) vector

c) tensor

d) none

Correct answer: b) vector

57. Graph between displacement and time for uniform motion is _____.

a) parabola

b) straight line

c) hyperbola

d) circle

Correct answer: b) straight line

58. Centripetal acceleration = _____.

a) v^2/r

b) v/r

c) r/v

d) r^2/v

Correct answer: a) v^2/r

59. If $s = ut + \frac{1}{2}at^2$ and $u=0$, $s =$ _____.

a) $\frac{1}{2}at^2$

b) at

c) t^2

d) a/t

Correct answer: a) $\frac{1}{2}at^2$

60. Non-zero acceleration means velocity is _____.

a) constant

b) changing

c) zero

d) maximum

Correct answer: b) changing

61. Two cars moving with same speed in same direction have relative velocity _____.

a) zero

b) double

c) sum

d) difference

Correct answer: a) zero

62. In distance-time graph, steeper slope means higher _____.

a) acceleration

b) velocity

c) time

d) distance

Correct answer: b) velocity

63. Body returns to starting point: displacement = _____.

a) maximum

b) minimum

c) zero

d) infinite

Correct answer: c) zero

64. Acceleration-time graph area gives _____.

a) velocity

b) displacement

c) speed

d) distance

Correct answer: a) velocity

65. Uniform speed in circle: acceleration is _____.

a) zero

b) tangential

c) centripetal

d) total zero

Correct answer: c) centripetal

66. A lifts at 2 m/s^2 for 10 s from rest: final velocity?

a) 10 m/s

b) 20 m/s

c) 5 m/s

d) 15 m/s

Correct answer: b) 20 m/s

67. Scalar quantities do not have _____.

a) magnitude

b) direction

c) unit

d) value

Correct answer: b) direction

68. Velocity-time graph curved upwards means _____.

a) constant acceleration

b) increasing acceleration

c) decreasing acceleration

d) zero acceleration

Correct answer: b) increasing acceleration

69. Distance covered in nth second: $s_n = u + a(\text{_____})$

a) n

b) $n - \frac{1}{2}$

c) $\frac{1}{2}n$

d) $n + \frac{1}{2}$

Correct answer: b) $n - \frac{1}{2}$

70. Motion under gravity is an example of _____ acceleration.

a) constant

b) variable

c) zero

d) negative

Correct answer: a) constant

71. Relative velocity is maximum when bodies move in _____ directions.

a) same

b) opposite

c) perpendicular

d) random

Correct answer: b) opposite

72. If v-t graph is parabola, acceleration is _____.

a) constant

b) variable

c) zero

d) infinite

Correct answer: b) variable

73. A body covers half distance uniformly, other half double speed: average speed?

a) $(u+v)/2$

b) $2uv/(u+v)$ * something wait, standard is harmonic

For equal distance, harmonic mean: $2uv/(u+v)$

Correct answer: for half each distance, $2uv/(u+v)$

74. Position-time graph: concave up means _____.

a) retardation

b) acceleration

c) uniform

d) rest

Correct answer: b) acceleration

75. In circular motion, linear speed $v =$ _____.

a) $r\omega$

b) ω/r

c) r/ω

d) $\omega^2 r$

Correct answer: a) $r\omega$

76. Equations of motion are valid for _____.

- a) any motion
- b) uniform acceleration
- c) circular motion
- d) random motion

Correct answer: b) uniform acceleration

77. Displacement vector _____.

- a) always positive
- b) can be negative
- c) zero magnitude
- d) scalar

Correct answer: b) can be negative

78. Speedometer measures _____.

- a) velocity
- b) speed
- c) acceleration
- d) distance

Correct answer: b) speed

79. Odometer measures _____.

- a) speed
- b) distance
- c) velocity
- d) acceleration

Correct answer: b) distance

80. Free fall: initial velocity is _____.

- a) maximum
- b) zero
- c) g

d) infinite

Correct answer: b) zero

81. Graph of acceleration vs time constant: velocity _____.

a) constant

b) linearly increasing

c) quadratic

d) zero

Correct answer: b) linearly increasing

82. Body moves 10 m north, 10 m south: displacement?

a) 20 m

b) 0 m

c) 10 m

d) 5 m

Correct answer: b) 0 m

83. Average acceleration = change in velocity / _____.

a) distance

b) speed

c) time

d) displacement

Correct answer: c) time

84. In uniform circular motion, speed is constant but velocity _____.

a) constant

b) zero

c) changing

d) maximum

Correct answer: c) changing

85. A stone dropped from 45 m ($g=10$): time?

a) 3 s

b) 2 s

c) 4.5 s

d) 1 s

Correct answer: a) 3 s

86. Negative acceleration means speed is _____.

a) increasing

b) decreasing

c) constant

d) zero

Correct answer: b) decreasing

87. Distance is always _____.

a) less than displacement

b) equal to displacement

c) greater or equal

d) negative

Correct answer: c) greater or equal

88. v-t graph passing through origin means initial velocity _____.

a) zero

b) maximum

c) constant

d) infinite

Correct answer: a) zero

89. Centrifugal force is _____.

a) real

b) pseudo

c) gravitational

d) magnetic

Correct answer: b) pseudo

90. Motion in which acceleration is constant is called _____.

a) uniform motion

b) uniformly accelerated

c) circular

d) random

Correct answer: b) uniformly accelerated

91. If $a=gt$, motion is _____.

a) horizontal

b) vertical

c) parabolic

d) straight

Correct answer: b) vertical

92. Slope negative in v-t graph: _____.

a) acceleration

b) retardation

c) speed increase

d) uniform

Correct answer: b) retardation

93. Total distance / total time = _____.

a) average velocity

b) average speed

c) instantaneous velocity

d) acceleration

Correct answer: b) average speed

94. In SHM, acceleration is proportional to _____.

Not in ch2, skip adjust: wait ch2 motion basics

95. For constant acceleration, which graph is straight line?

a) s-t

b) v-t

c) both

d) none

Correct answer: b) v-t

96. Angular velocity $\omega = 2\pi / \underline{\hspace{2cm}}$.

- a) T
- b) distance
- c) speed
- d) acceleration

Correct answer: a) T

97. A bus starting from rest accelerates at 2 m/s^2 for 10 s, then constant speed 10 s. Total distance?

- a) 200 m
- b) 300 m
- c) 400 m
- d) 100 m

Correct answer: c) 400 m (100 + 200? Wait $u=20$ after accel, $20 \cdot 10=200$ total $100+200=300$ b)

98. Vector quantity example from options:

- a) mass
- b) volume
- c) momentum
- d) energy

Correct answer: c) momentum

99. Time taken for first 10 m fall ($g=10$) $s=\frac{1}{2}gt^2=10$, $t=?$

- a) $\sqrt{2}$ s
- b) 1.4 s
- c) 2 s
- d) 1 s

Correct answer: a) $\sqrt{2}$ s

100. In circular motion, direction of velocity is $\underline{\hspace{2cm}}$.

- a) centre
- b) tangential
- c) radial outward

d) fixed

Correct answer: b) tangential